

<Q> The formal charges for the carbon and nitrogen atoms in $(\text{CH}_3)_4\text{N}$ respectively?

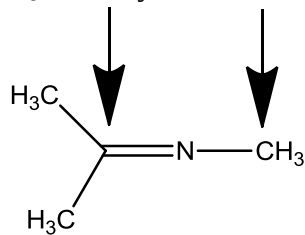
<A> 0, 0

<B+> 0, +1

<C> +1, 0

<D> +4, +3

<Q> The hybridization of the indicated atoms (from left to right) are?



<A> sp^2 , sp^2

 sp^3 , sp^3

<C> sp^3 , sp^2

<D+> sp^2 , sp^3

<Q> The most polar bond is?

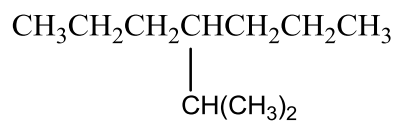
<A> NaCl

<B+> LiF

<C> KBr

<D> NH_3

<Q> The IUPAC name for the following compound is



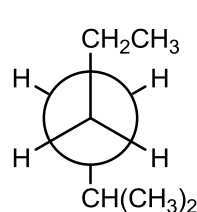
<A> 4-Isopropyloctane

<B+> 4-Isopropylheptane

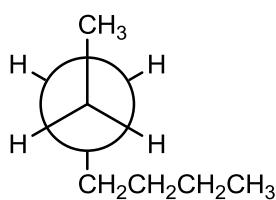
<C> 2-Methyl-3-propylhexane

<D> 2-Methyl-3-propylheptane

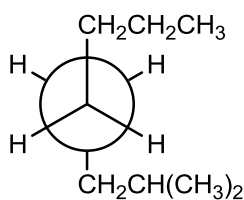
<Q> For rotation about C3-C4 bond of 2-methylhexane, the most stable conformer is



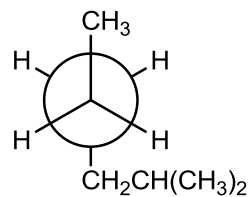
I



II



III



IV

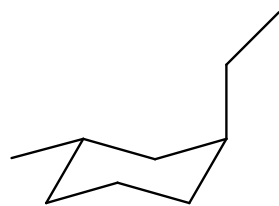
<A+> I

 II

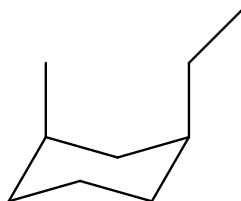
<C> III

<D> IV

<Q> The most stable conformation of *trans*-1-ethyl-2-methylcyclohexane is:



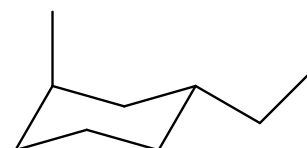
I



II



III



IV

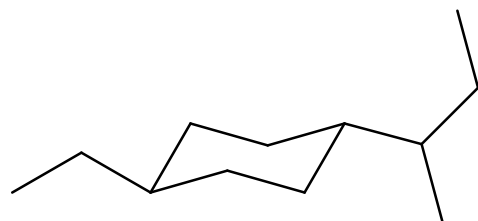
<A> I

 II

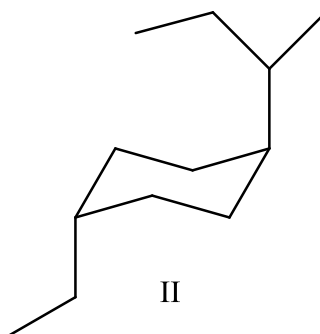
<C> III

<D+> IV

<Q> The relationship between I and II is



I



II

<A> constitutional isomers

<B+> conformational isomers

<C> Geometric isomer

<D> different compounds (not isomers)

<Q> The most typical reaction of simple alkenes is ...

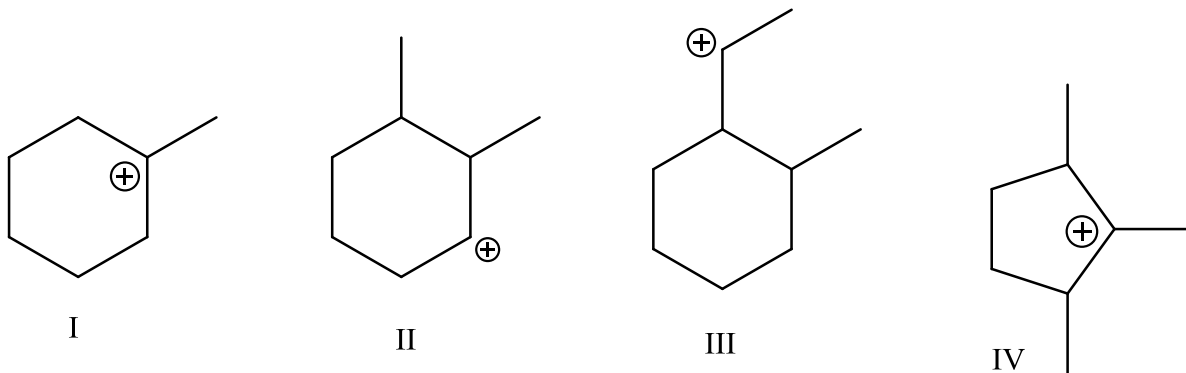
<A> electrophilic substitution

 nucleophilic substitution

<C+> electrophilic addition

<D> nucleophilic addition

<Q> The structure that represent 3° carbocation



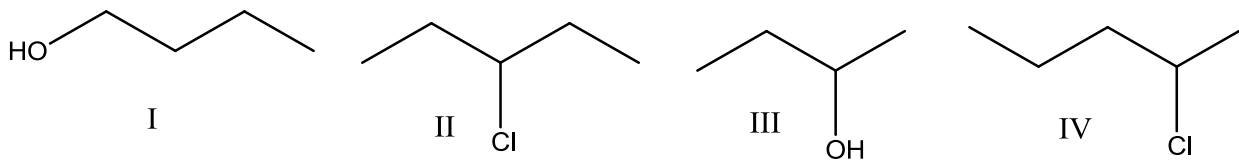
<A> I only

<B+> I and IV

<C> III only

<D> II and III

<Q> Consider the following structures, the product from the reaction of 2-pentene + HCl is.....



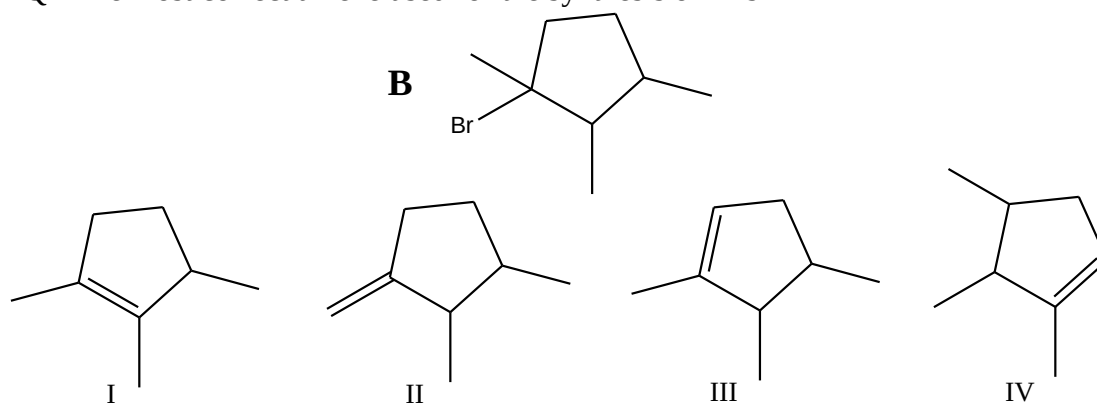
<A> II only

 IV only

<C+> II and IV

<D> II and III

<Q> The most correct alkene used for the synthesis of B is



<A> I only

 II and III

<C+> II, III and IV

<D> III only

<Q> How many alkenes can be hydrogenated to give pentane:

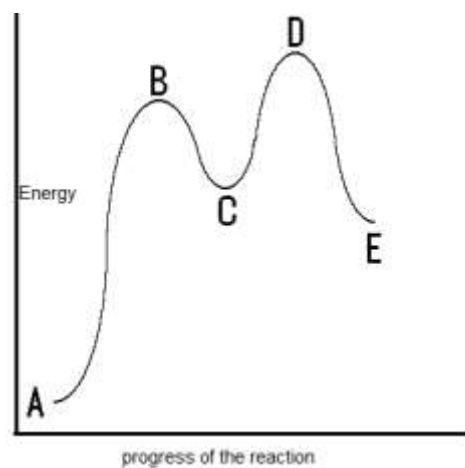
<A> 2

 3

<C> 4

<D+> 5

<Q> Consider the following reaction coordinate diagram, the most stable transition state is



<A> A

<B+> B

<C> C

<D> D

<Q> Consider the above reaction coordinate diagram, the favorable step is

<A> A to B

 A to C

<C> C to D

<D+> C to E

<Q> Consider the above reaction coordinate diagram, the reactant in the slowest step is

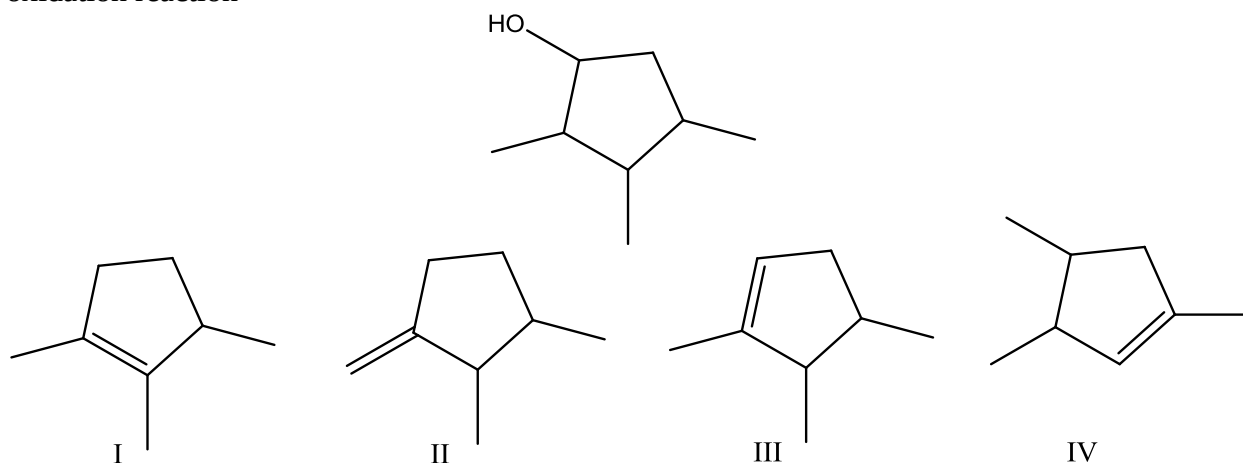
<A+> A

 B

<C> C

<D> D

<Q> Which of the following alkene will give only the given alcohol upon hydroboration-oxidation reaction



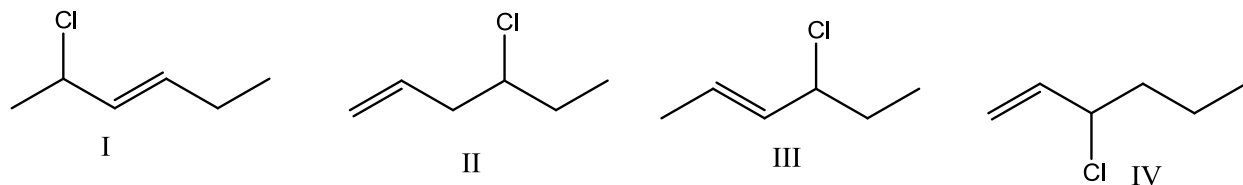
<A> I

 II and IV

<C+> III

<D> III and IV

<Q> The possible products when adding 1 mole of hydrogen chloride (HCl) to 1,3-hexadiene



<A> I and III

 II and IV

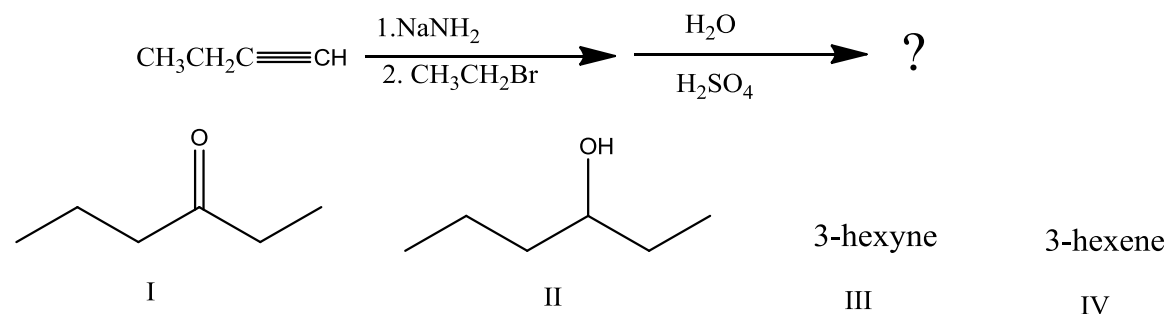
<C> I, III and IV

<D+> I, II, III and IV

<Q> Which of the following compounds is the most acidic?

- A) 2-heptyne B) 4-methyl-2-heptyne C) 1-heptyne D) 2-butyne

<Q> What is the final product of the following reaction in the following sequence?



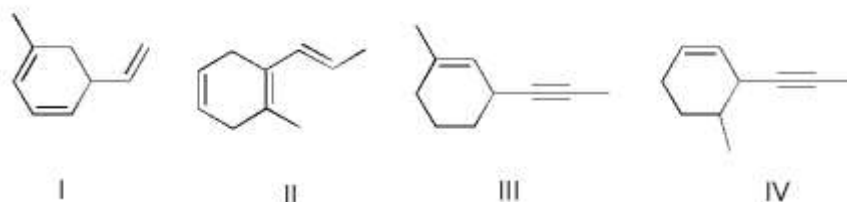
<A+> I

 II

<C> III

<D> IV

<Q> Which of the following will give 1-methyl-3-propylcyclohexane upon its reaction with excess H_2 in the presence of Pt catalyst?



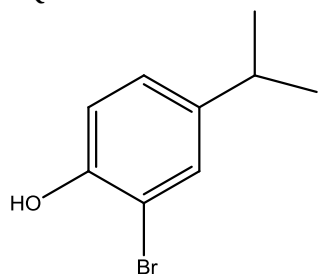
<A+> III only

 II and IV

<C> II and III

<D> II

<Q> The suitable name for



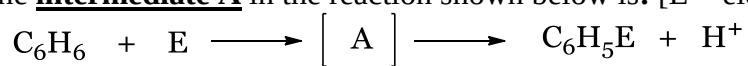
<A+> 2-bromo-4-isopropylphenol

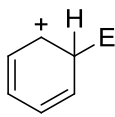
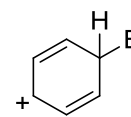
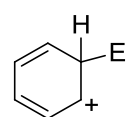
 2-isopropyl-4-bromophenol

<C> 2-bromo-4-tert-butylphenol

<D> 3-bromo-1-isopropylphenol

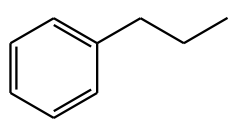
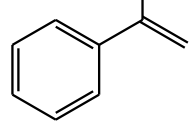
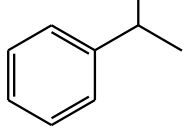
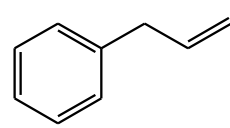
<Q> The **intermediate A** in the reaction shown below is: [E = electrophile]



- A)  B)  C)  D) All of these

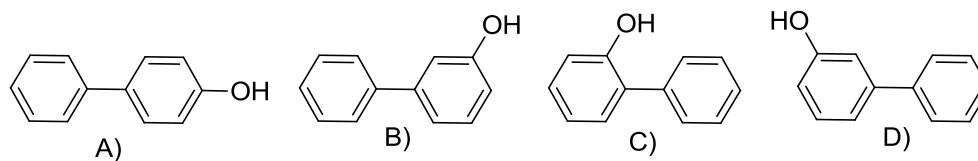
D

<Q> When benzene is treated with propene and sulfuric acid, the major product is:

- A)  B)  C)  D) 

C

<Q> The structure of *ortho*-phenylphenol is:



C

<Q> Which of the following is an electrophile in aromatic ring nitration?

- A) N_2O^+ B) NO^+ C) N_2^+ D) NO_2^+

D